

datalex



Cloud Computing



Alexandre BERGERE

Head of Data & AI Engineer & Partners at DataGalaxy, Data Architect independent
Delta & openLineage lover

Data Expert
(Audit, Modeling, Urbanization, BI's project – Azure, Big Data, Spark, openLineage & Delta lover)

Trainer
(BI, Big Data, MongoDB, NoSQL, SQL, Cloud Computing)

Speaker
(Pass Summit 2018, Spark AI Tour 2020, …)

Cloud & data tech articles
(medium, LinkedIn, slideshare)

Avanade
Sr Anls, Data Analyst

2016 - 2019

Trainer
2016 - x

ESAIP, ESME, ECS, NEOMA, ESEO, EFREI, Sorbonne Paris

BI, Big Data, MongoDB, NoSQL, Panorama de la data, Cloud computing

OpenClassroom
Mentor

2019 - x

Datalex
2019 - x

Data Architect Freelance

DataRedKite
CTO & Co-Founder

2020 - 2023

DataGalaxy
2021 - x

Head of Data & AI Engineer

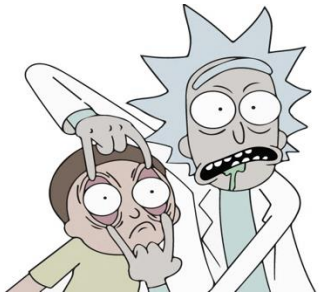
Don’t get lost



Practice



Demo

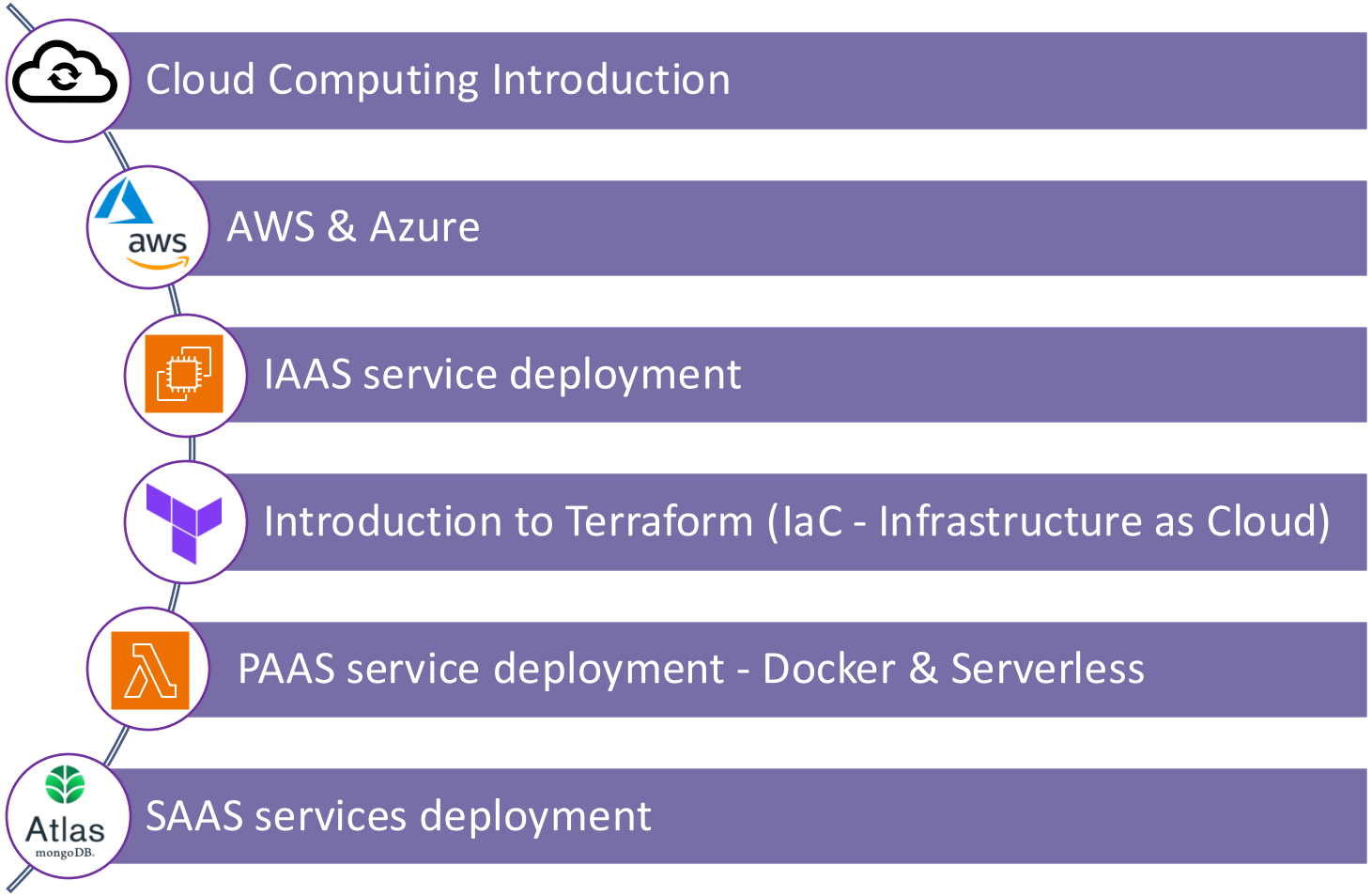


Important



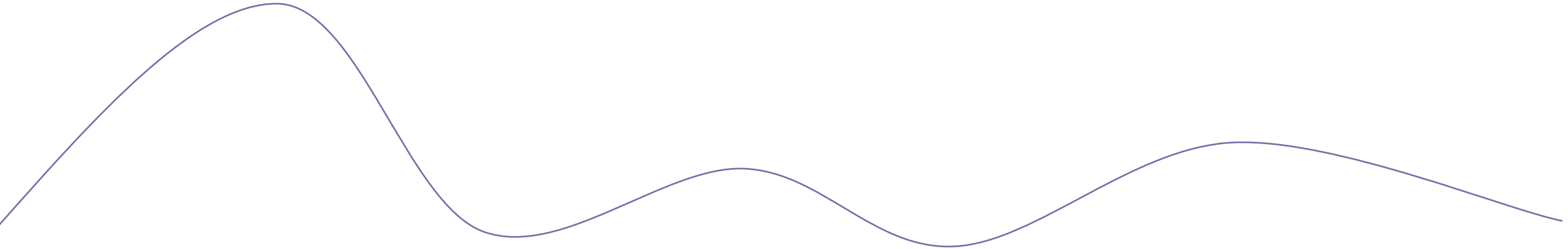
Tips

Summary


Cloud Computing Introduction
aws
AWS & Azure
IAAS service deployment
Introduction to Terraform (IaC - Infrastructure as Cloud)
PAAS service deployment - Docker & Serverless
Atlas
mongoDB.
SAAS services deployment

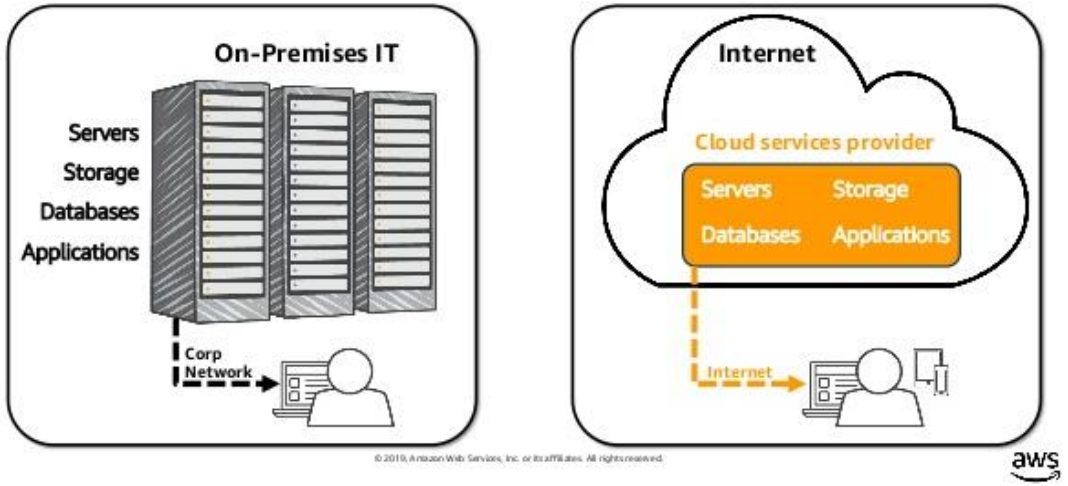
Cloud Computing

Basics concepts



What is Cloud Computing?

Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources through a cloud services platform via the internet with pay-as-you-go pricing.



https://aws.amazon.com/what-is-cloud-computing/?nc1=h_ls

On-Premise vs Cloud

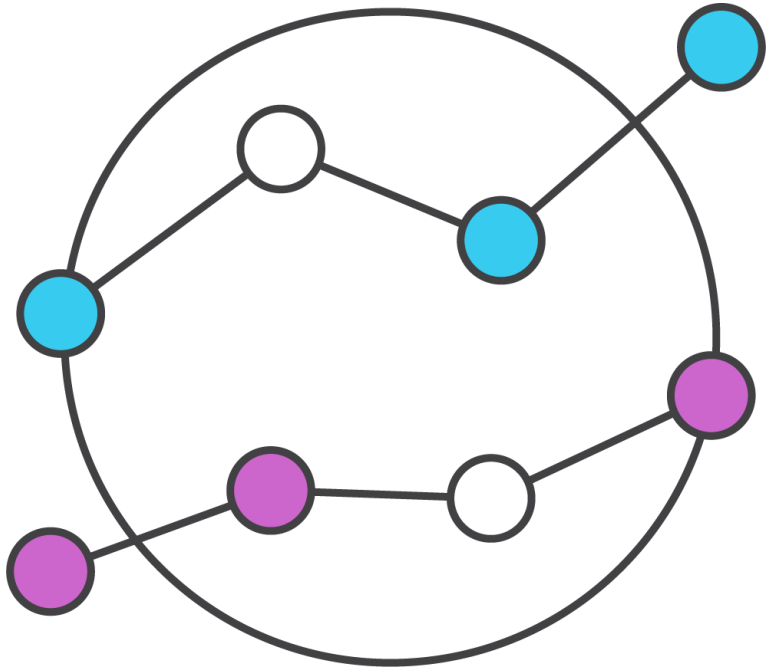
ON-PREMISE	FEATURES	CLOUD
The organization owns and manages its servers, networking, and data centers.	Infrastructure Ownership	Infrastructure is provided and maintained by a third-party cloud service provider.
Scaling requires physical procurement and setup of additional hardware.	Scalability	Easily scalable, resources can be provisioned or de-provisioned on demand.
Capital expenditure (CapEx) for upfront hardware costs.	Cost Model	Operational expenditure (OpEx) model with pay-as-you-go pricing.
Offers greater control and customization over infrastructure.	Flexibility	Provides flexibility to quickly adapt and scale resources based on demand.
Full control and responsibility for maintenance, security, and updates.	Responsibility	Shared responsibility with the cloud provider handling infrastructure maintenance.
Security measures are implemented and managed by the organization.	Security	Cloud providers implement robust security measures, but shared responsibility requires organizations to manage certain aspects.
Deployment can be time-consuming due to hardware procurement and setup.	Time to Deploy	Rapid deployment with resources provisioned within minutes.
Limited to the physical location of data centers.	Geographical Reach	Globally distributed data centers for widespread accessibility.
Organizations are responsible for updates and maintenance.	Updates and Maintenance	Cloud providers handle updates, patches, and maintenance (depending of the types of cloud services).
Organization responsible for implementing disaster recovery plans.	Disaster Recovery	Cloud providers offer built-in disaster recovery options and backup services.

Why cloud computing

The Expense Advantage

No need to buy hardware or build data centers.

- Pay as you consume resources.
- Reduce upfront capital costs.

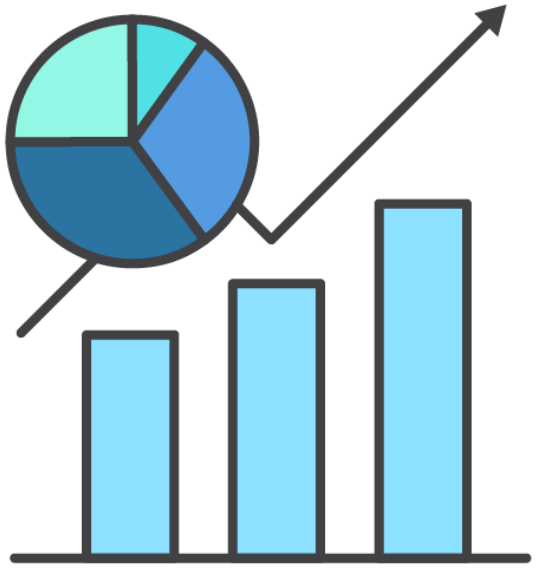


Why cloud computing

The Scale Advantage

Leverage large economies of scale.

- Lower cost than on your own
  - Specialized hardware and software
  - Large volume hardware purchasing

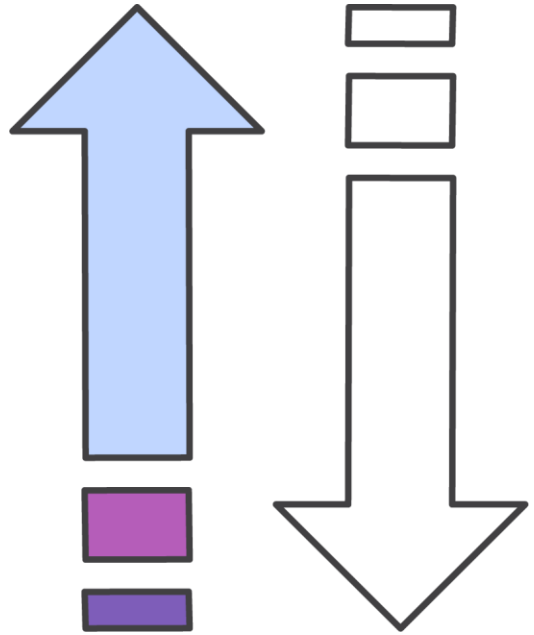


Why cloud computing

The Capacity Advantage

Stop guessing about capacity.

- Scale up and down as needed.
- Don’t need to overprovision.

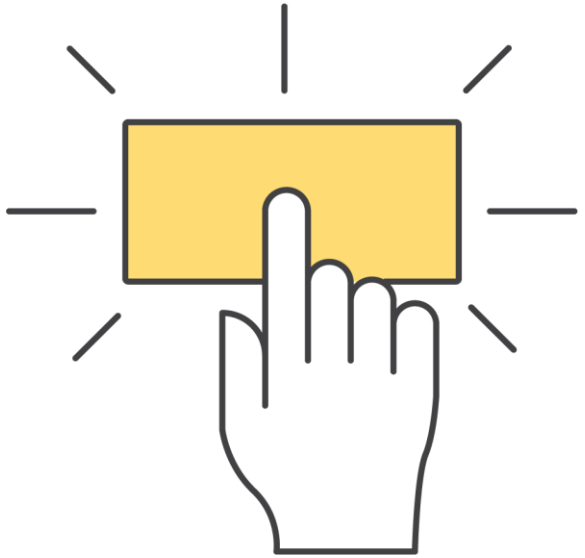


Why cloud computing

The Speed Advantage

No need to wait for hardware to be delivered and set up.

- New IT resources are just a click away.
- Reduce resource development time.



Why cloud computing

The Focus Advantage

Focus on your applications, not your infrastructure.

- Free up resources to spend on new projects.
- Stop spending money running and maintaining data centers.
- Disposable resources allow for quick experimentation.



Why cloud computing

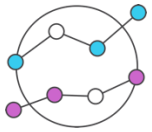
The Global Advantage

Go global in minutes.

- Multiple AWS Regions around the world
- Keep your application close to your users
- Facilitate high availability and disaster recovery



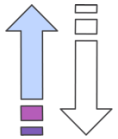
Why cloud computing



- Pay only for the amount you consume (stop guessing capacity)



- Benefit from massive economies of scale (Stop spending money on running and maintaining data centers)



- Stop guessing about capacity



- Increase speed and agility (minutes to waiting ressources)



- Focus on what matters



- Go global in minutes

Market’s leaders











Market leader

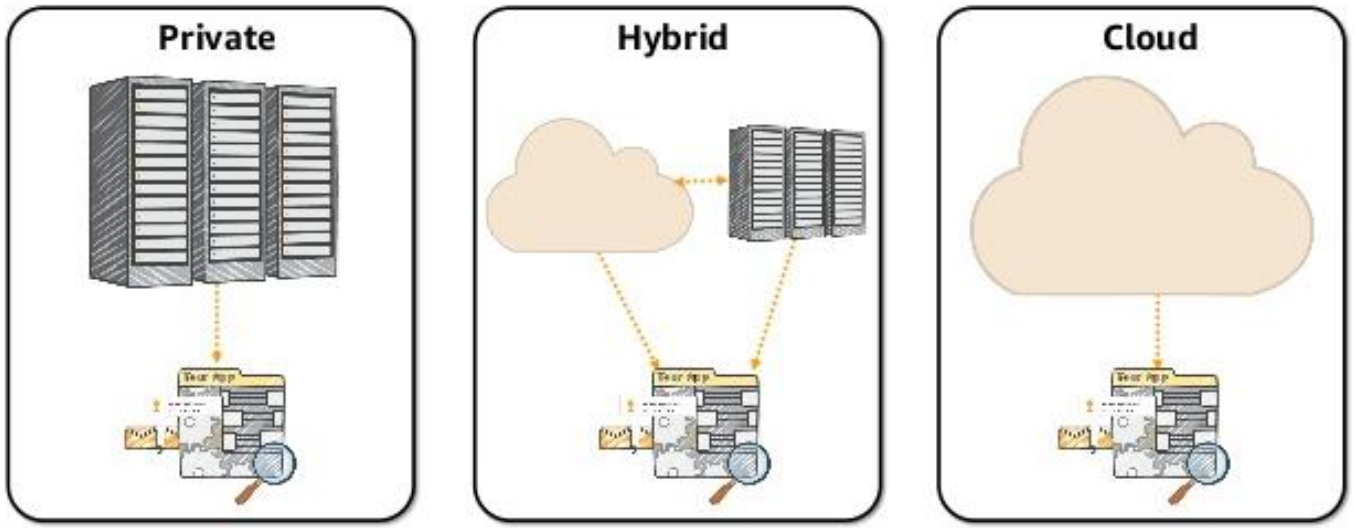
Figure 1. Magic Quadrant for Cloud Infrastructure and Platform Services



Gartner

Features	Amazon Web Services	Google Cloud	Microsoft Azure
Market Share 2019	32.3%	5.8%	16.9%
Annual Growth	36%	87.8%	63.9%
Regions	24	22	58
Countries served	245	200+	100+
Number of services	212	60+	100+

Types of cloud deployment models



Cloud deployment models

There are three different cloud deployment models. A cloud deployment model defines where your data is stored and how your customers interact with it – how do they get to it, and where do the applications run? It also depends on how much of your own infrastructure you want or need to manage:

- Public cloud
- Private cloud
- Hybrid cloud

Public cloud



Advantages:

- High scalability/agility – you don’t have to buy a new server in order to scale
- Pay-as-you-go pricing – you pay only for what you use, no CapEx costs
- You’re not responsible for maintenance or updates of the hardware
- Minimal technical knowledge to set up and use - you can leverage the skills and expertise of the cloud provider to ensure workloads are secure, safe, and highly available

Disadvantages:

- There may be specific security requirements that cannot be met by using public cloud
- There may be government policies, industry standards, or legal requirements which public clouds cannot meet
- You don't own the hardware or services and cannot manage them as you may want to
- Unique business requirements, such as having to maintain a legacy application might be hard to meet

Private cloud



Advantages:

- You can ensure the configuration can support any scenario or legacy application
- You can control (and responsibility) over security
- Private clouds can meet strict security, compliance, or legal requirements

Disadvantages:

- You have some initial CapEx costs and must purchase the hardware for startup and maintenance
- Owning the equipment limits the agility - to scale you must buy, install, and setup new hardware
- Private clouds require IT skills and expertise that's hard to come by

Hybrid cloud

A hybrid cloud combines public and private clouds, allowing you to run your applications in the most appropriate location. For example, you could host a website in the public cloud and link it to a highly secure database hosted in your private cloud (or on-premises datacenter).

Advantages:

- You can keep any systems running and accessible that use out-of-date hardware or an out-of-date operating system
- You have flexibility with what you run locally versus in the cloud
- You can take advantage of economies of scale from public cloud providers for services and resources where it's cheaper, and then supplement with your own equipment when it's not
- You can use your own equipment to meet security, compliance, or legacy scenarios where you need to completely control the environment

Disadvantages:

- It can be more expensive than selecting one deployment model since it involves some CapEx cost up front
- It can be more complicated to set up and manage

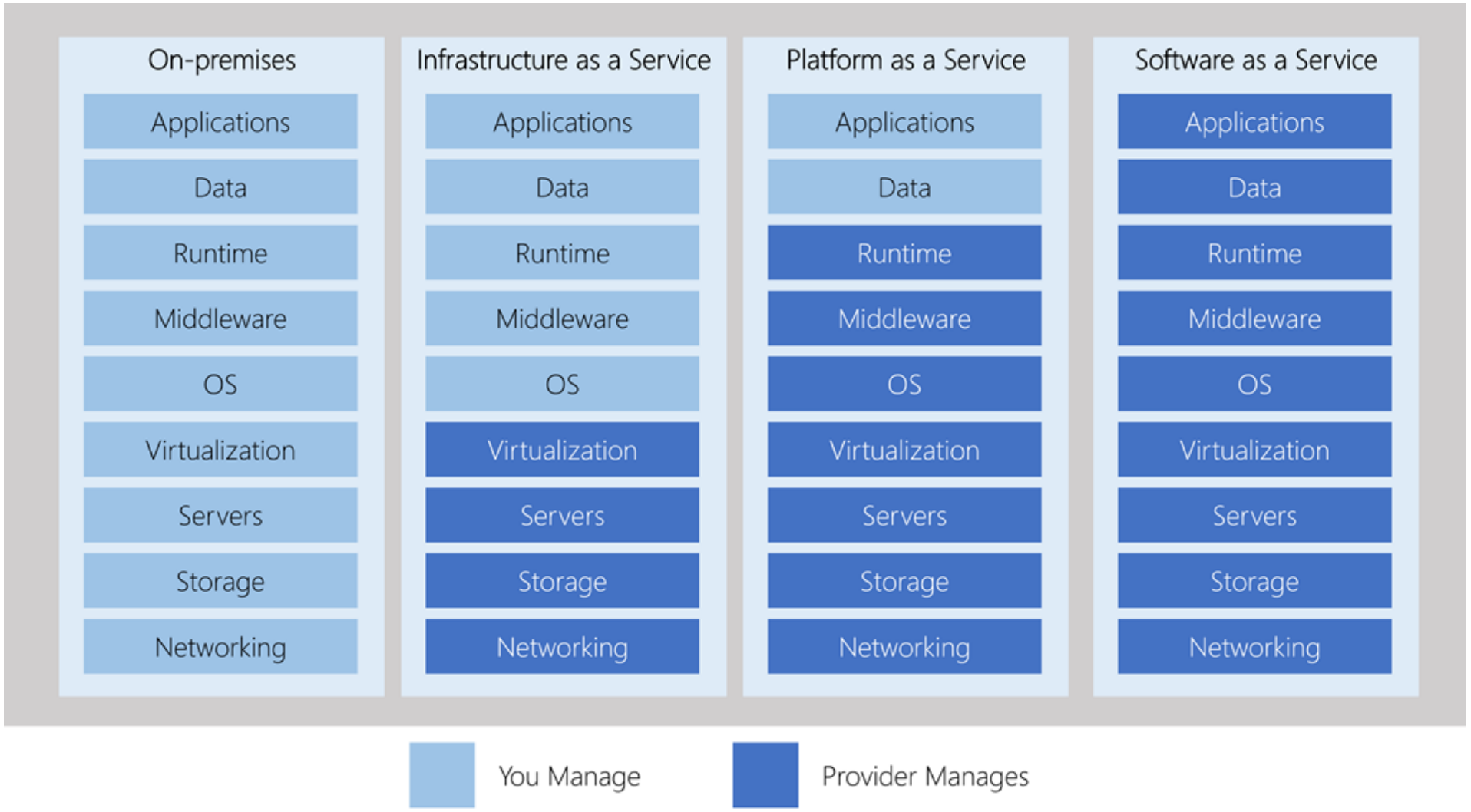
Types of cloud services



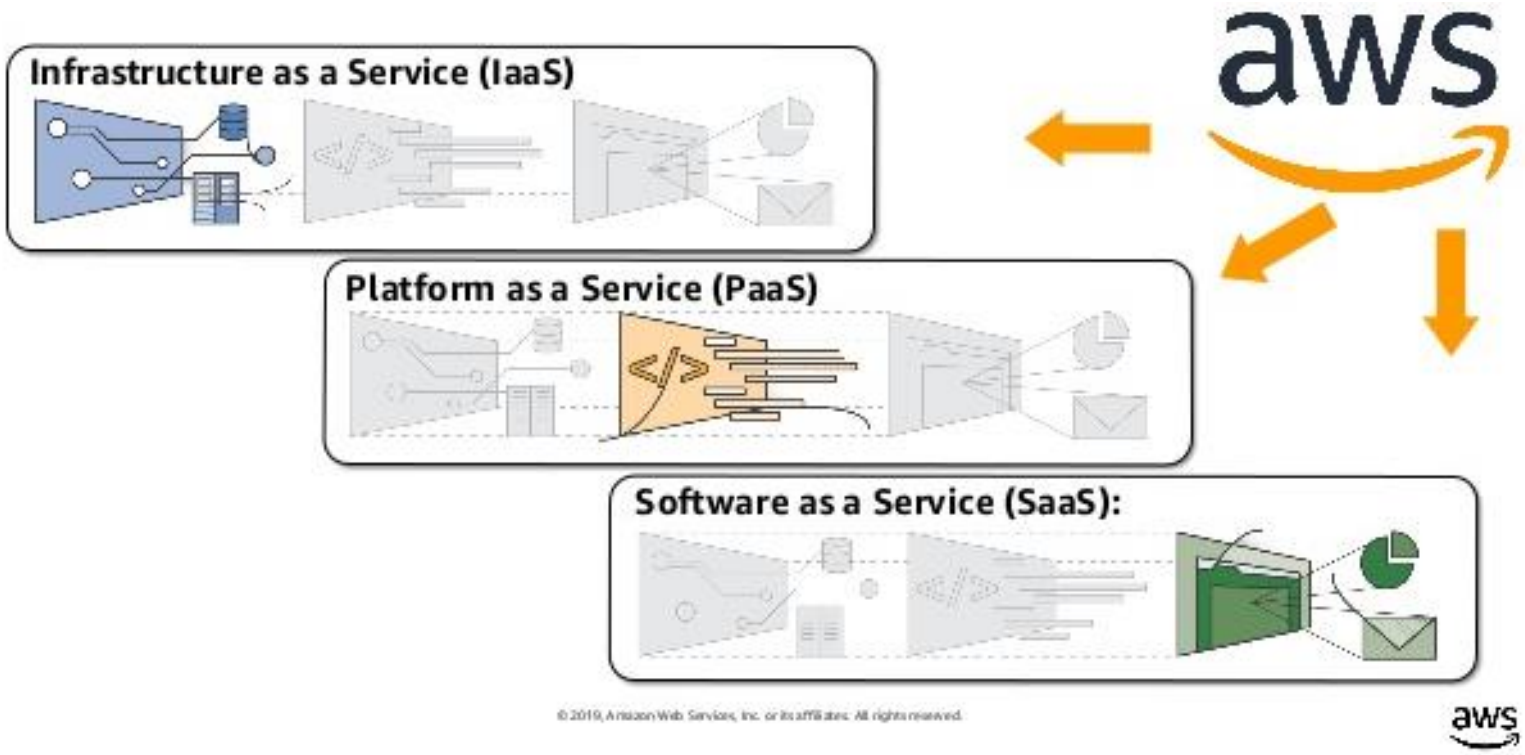




Management responsibilities



Management responsibilities



Infrastructure as a service (IaaS)

IaaS is commonly used in the following scenarios:

- Migrating workloads. Typically, IaaS facilities are managed in a similar way as on-premises infrastructure and provide an easy migration path for moving existing applications to the cloud.
- Test and development. Teams can quickly set up and dismantle test and development environments, bringing new applications to market faster. IaaS makes scaling development and testing environments, fast and economical.
- Website hosting. Running websites using IaaS can be less expensive compared to traditional web hosting.
- Storage, backup, and recovery. Organizations avoid the capital outlay and complexity of storage management, which typically requires skilled staff to manage data and meet legal and compliance requirements. IaaS is useful for managing unpredictable demand and steadily growing storage needs. IaaS can also simplify the planning and management of backup and recovery systems.

When using IaaS, ensuring that a service is up and running is a shared responsibility: the cloud provider is responsible for ensuring the cloud infrastructure is functioning correctly; the cloud customer is responsible for ensuring the service they are using is configured correctly, is up to date, and is available to their customers. This is referred to as the shared responsibility model.

Platform as a service (PaaS)

PaaS provides an environment for building, testing, and deploying software applications. The goal of PaaS is to help you create an application quickly without managing the underlying infrastructure. For example, when deploying a web application using PaaS, you don't have to install an operating system, web server, or even system updates.

PaaS is a complete development and deployment environment in the cloud, with resources that enable organizations to deliver everything from simple cloud-based apps to sophisticated cloud-enabled enterprise applications. Resources are purchased from a cloud service provider on a pay-as-you-go basis and accessed over a secure Internet connection.

PaaS is commonly used in the following scenarios:

- Development framework. PaaS provides a framework that developers can build upon to develop or customize cloud-based applications. Just like Microsoft Excel macro, PaaS lets developers create applications using built-in software components. Cloud features such as scalability, high-availability, and multi-tenant capability are included, reducing the amount of coding that developers must do.

- Analytics or business intelligence. Tools provided as a service with PaaS allow organizations to analyze and mine their data. They can find insights and patterns, and predict outcomes to improve business decisions such as forecasting, product design, and investment returns.

Software as a service (SaaS)

SaaS is software that is centrally hosted and managed for the end customer. It is usually based on an architecture where one version of the application is used for all customers, and licensed through a monthly or annual subscription. Office 365, Skype, and Dynamics CRM Online are perfect examples of SaaS software.

Cost and Ownership

	IaaS	PaaS	SaaS
Upfront costs	There are no upfront costs. Users pay only for what they consume.	There are no upfront costs. Users pay only for what they consume.	Users have no upfront costs; they pay a subscription, typically on a monthly or annual basis.
User ownership	The user is responsible for the purchase, installation, configuration, and management of their own software, operating systems, middleware, and applications.	The user is responsible for the development of their own applications. However, they are not responsible for managing the server or infrastructure. This allows the user to focus on the application or workload they want to run.	Users just use the application software; they are not responsible for any maintenance or management of that software.
Cloud provider ownership	The cloud provider is responsible for ensuring that the underlying cloud infrastructure (such as virtual machines, storage, and networking) is available for the user.	The cloud provider is responsible for operating system management, network, and service configuration. Cloud providers are typically responsible for everything apart from the application that a user wants to run. They provide a complete managed platform on which to run the application.	The cloud provider is responsible for the provision, management, and maintenance of the application software.

Management responsibilities – AWS example





Amazon EC2

Provision virtual cloud servers on Amazon EC2 with operating systems pre-installed on them. You have to install the application code and related software on the virtual machine.





AWS Lambda

Run application code without provisioning or managing infrastructure. For instance, you can simply write and upload the application code as a .zip file.





Amazon Chime

Amazon Chime is a communications service that lets you meet, chat, and place business calls inside and outside your organization, all using a single application.

Management responsibilities – Azure example





Azure virtual machine

Provision virtual cloud servers on Azure Virtual Machine with operating systems pre-installed on them. You have to install the application code and related software on the virtual machine.





Azure SQL Database

Run application code without provisioning or managing infrastructure. For instance, you can simply write and upload the application code as a .zip file.

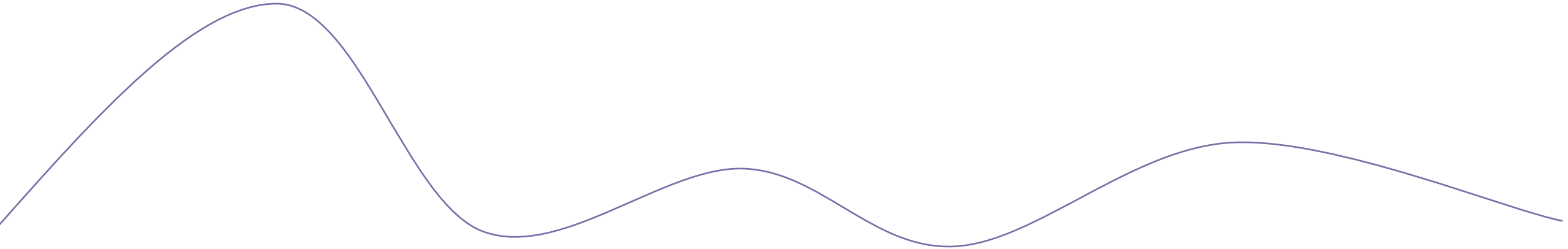




Office 365

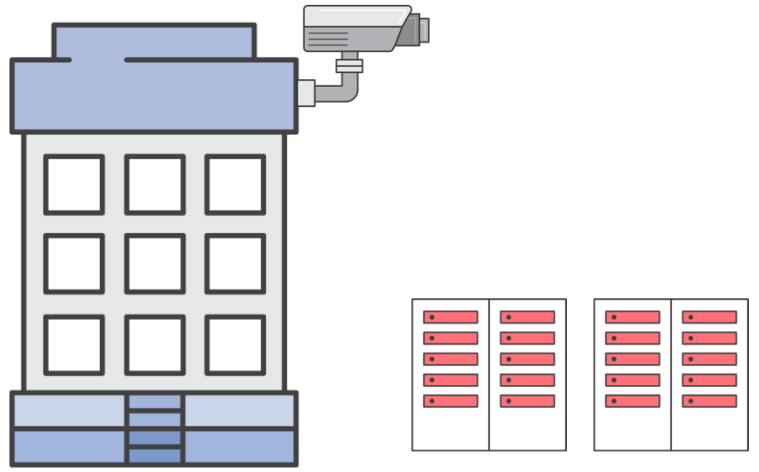
Office 365 is a subscription-based version of the Microsoft Office Suite, and you have a few options when purchasing an account.

Scalable & Available



Data Centers

- A single data center typically houses tens of thousands of servers
- All data centers are online, not “cold”
- AWS / Azure custom network equipment:
  - Multi-ODM (Original design manufacturers) sourced
  - Customized network protocol stack



Choose your location



Choose your location



What is a region?

A region is a geographical area on the planet containing at least one, but potentially multiple datacenters that are nearby and networked together with a low-latency network. Clouder assigns and controls the resources within each region to ensure workloads are appropriately balanced.

Choose your region



Latency

If your application is sensitive to latency (the delay between a request for data and the response), choose a Region that is close to your user base. This helps prevent long wait times for your customers. Synchronous applications such as gaming, telephony, WebSockets, and Internet of Things (IoT) are significantly affected by high latency. Asynchronous workloads, such as ecommerce applications, can also suffer from user connectivity delays.

Due to the local economy and the physical nature of operating data centers, prices vary from one Region to another. Internet connectivity, imported equipment costs, customs, real estate, and other factors impact a Region's pricing. Instead of charging a flat rate worldwide, AWS charges based on the financial factors specific to each Region.



Pricing



Service Availability

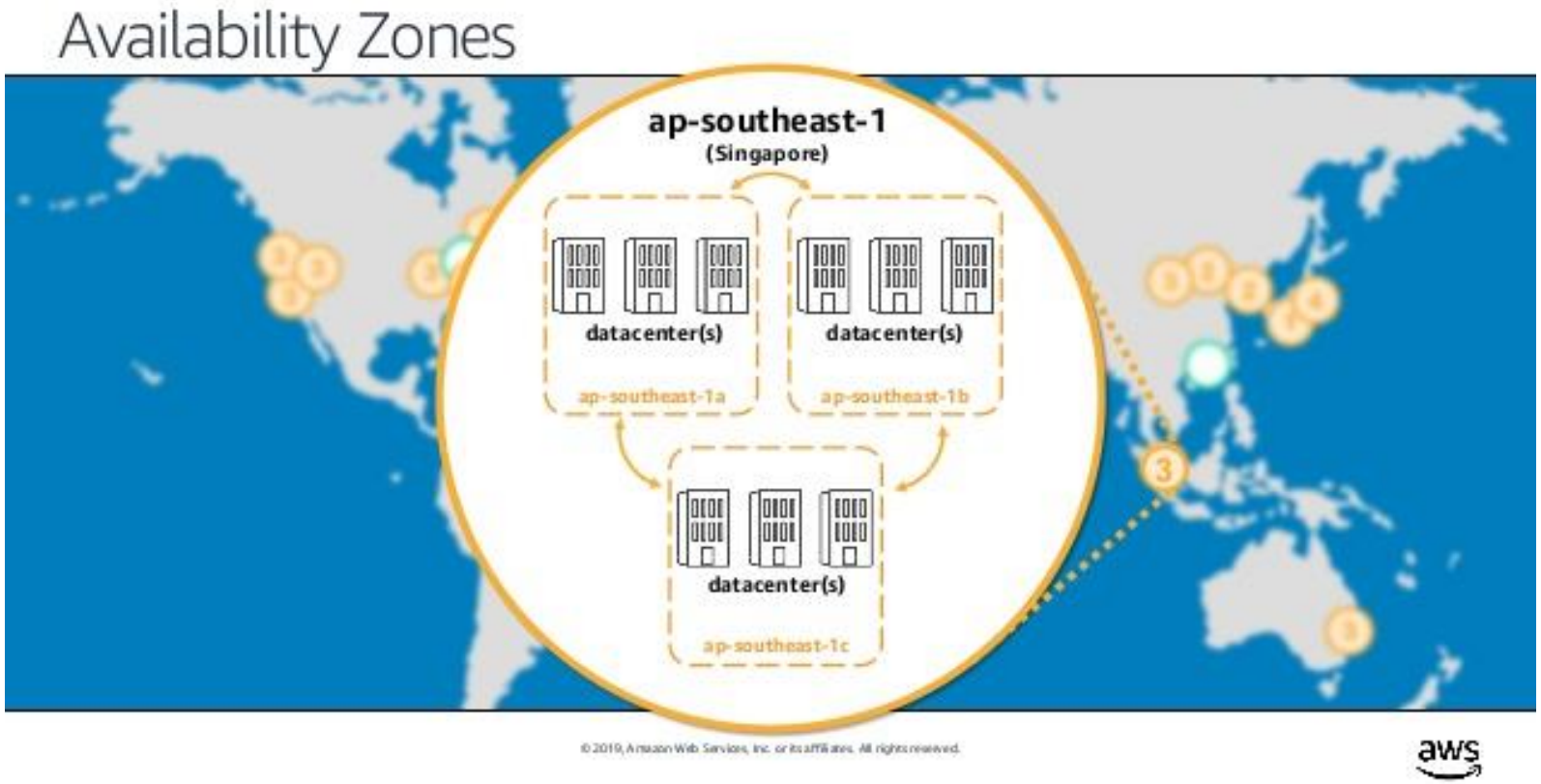
Some services might not be available in some Regions. The AWS documentation provides a table that shows the services available in each Region.

Enterprise companies often must comply with regulations that require customer data to be stored in a specific geographic territory. If applicable, choose a Region that meets your compliance requirements.



Data Compliance

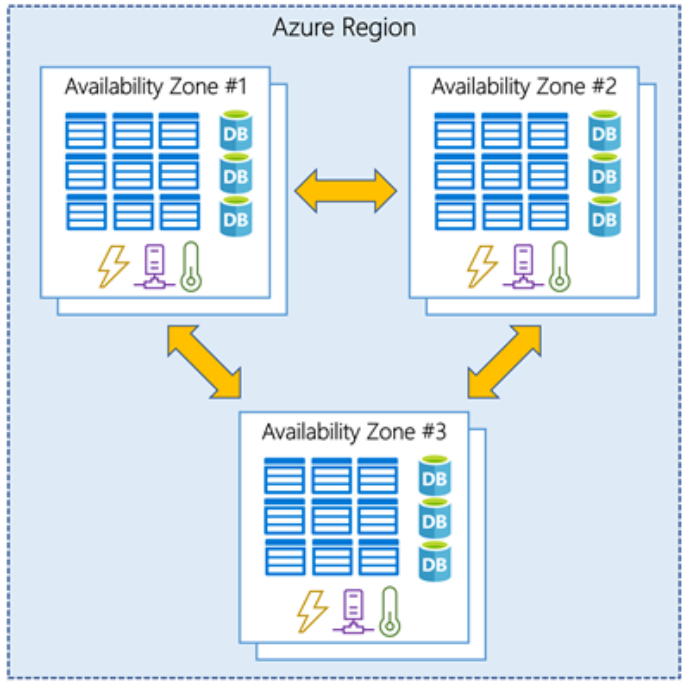
Avaibility zones



Availability Zones

Availability Zones are physically separate datacenters within a region.

Each Availability Zone is made up of one or more datacenters equipped with independent power, cooling, and networking. It is set up to be an isolation boundary. If one zone goes down, the other continues working. Availability Zones are connected through high-speed, private fiber-optic networks.



Supported regions

Not every region has support for Availability Zones.

https://docs.microsoft.com/en-us/learn/modules/explore-azure-infrastructure/4-availability-zones

Avaibility zones

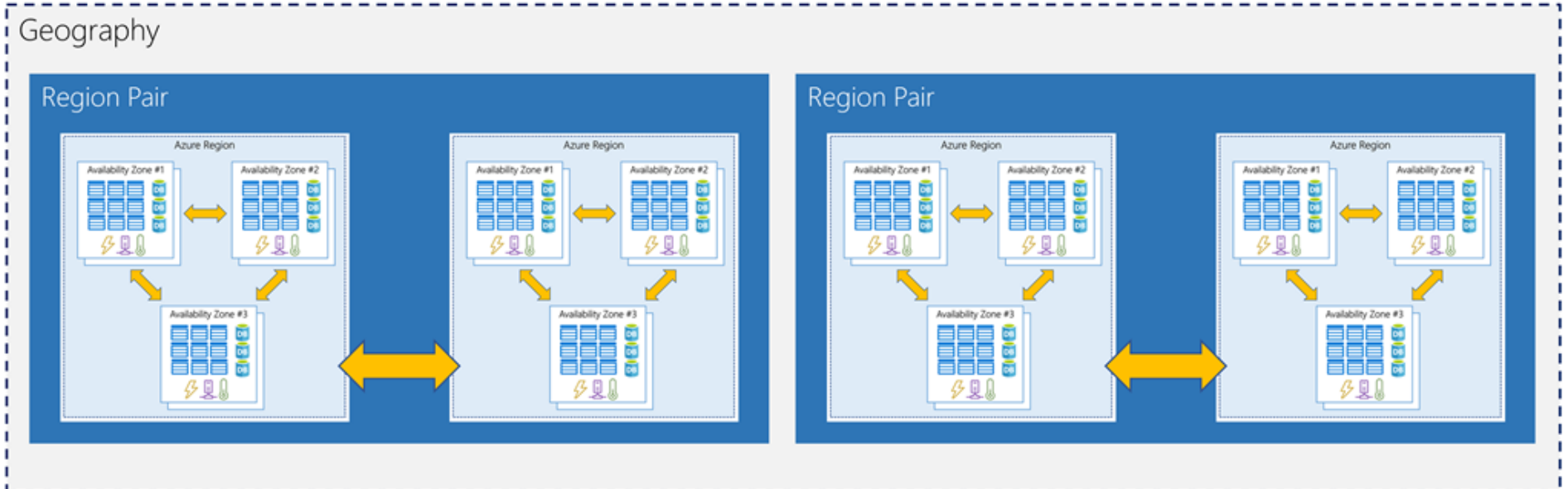
Cloud Native Architecture Best Practice: Disperse workload across multiple zones within a region to make the stack highly available and more durable against hardware or application component failure.

This often costs nothing extra to implement, but gv es stacks a distinct operational advantage by dispersing ciloyte nodes across multiple isolated data centers.

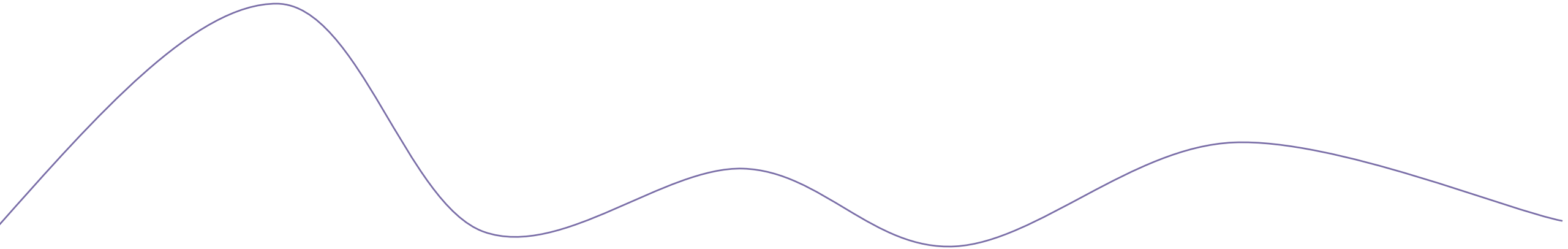
Region Pairs

Availability zones are created using one or more datacenters, and there are a minimum of three zones within a single region. However, it's possible that a large enough disaster could cause an outage big enough to affect even two datacenters. That's why Azure also creates region pairs.

Each Azure region can be paired with another region within the same geography (such as US, Europe, or Asia) at least 300 miles away. This approach allows for the **replication of resources** (such as virtual machine storage) across a geography that helps reduce the likelihood of interruptions due to events such as natural disasters, civil unrest, power outages, or physical network outages affecting both regions at once.

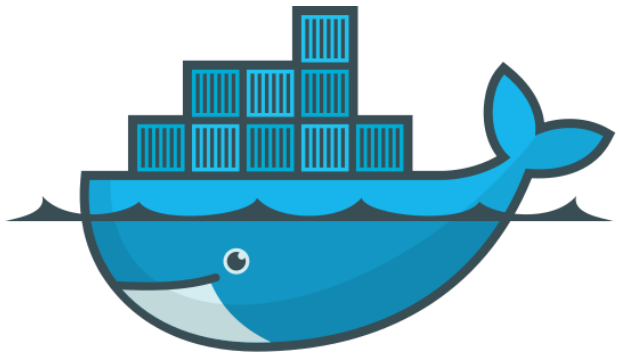


Cloud-native toolkit



Docker

https://docs.docker.com/get-started/overview/



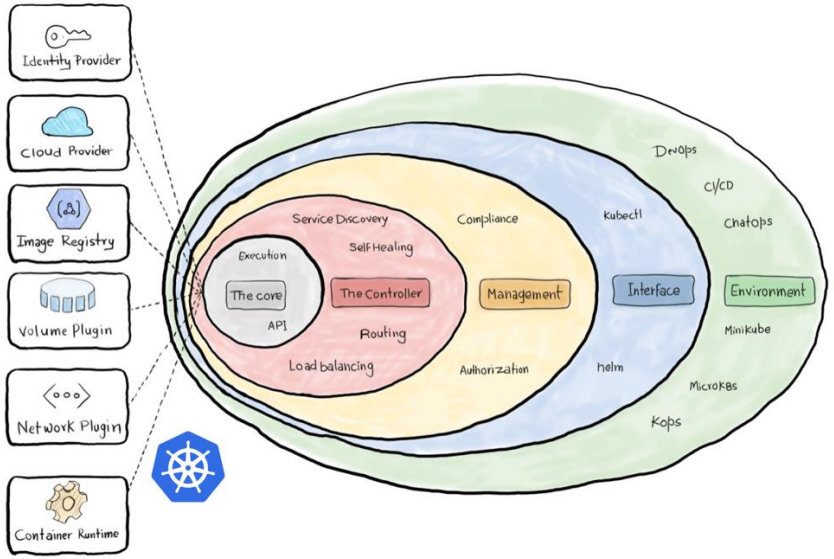
https://github.com/docker

Docker is an open platform for developing, shipping, and running applications. Docker enables you to separate your applications from your infrastructure so you can deliver software quickly.

With Docker, you can manage your infrastructure in the same ways you manage your applications.



Reading: Concepts and Why It Matters



Kubernetes

https://docs.docker.com/get-started/overview/



https://github.com/kubernetes

Kubernetes, also known as K8s, is an open source container orchestration engine for automating deployment, scaling, and management of containerized applications.

The open source project is hosted by the Cloud Native Computing Foundation (CNCF).



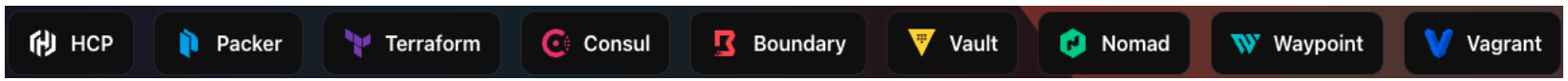
Training: kubernetes-the-hard-way.

Terraform

https://developer.hashicorp.com/terraform/docs

Terraform is an infrastructure as code tool that lets you build, change, and version infrastructure safely and efficiently.

This includes low-level components like compute instances, storage, and networking, as well as high-level components like DNS entries and SaaS features.



https://github.com/hashicorp/terraform

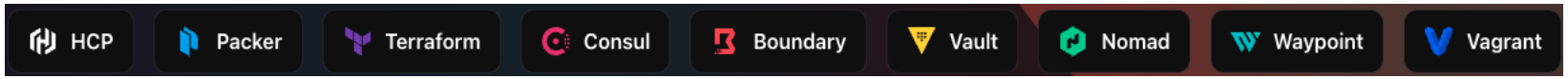
Vault

https://developer.hashicorp.com/vault/docs

https://github.com/hashicorp/vault

Vault is an identity-based secrets and encryption management system.

A secret is anything that you want to tightly control access to, such as API encryption keys, passwords, and certificates. Vault provides encryption services that are gated by authentication and authorization methods



Chaos monkey

https://netflix.github.io/chaosmonkey/

Chaos Monkey randomly terminates virtual machine instances and containers that run inside of your production environment. Exposing engineers to failures more frequently incentivizes them to build resilient services.

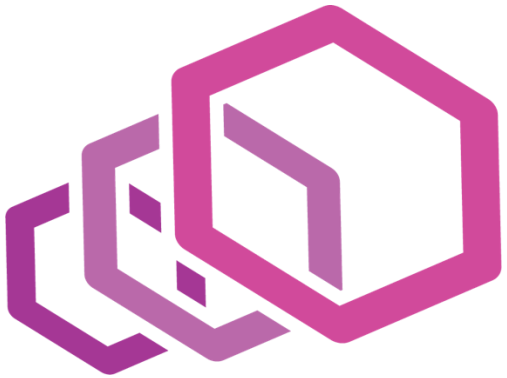


Originally developed at Netflix, Chaos Monkey is an example of a tool that follows the Principles of Chaos Engineering.

https://github.com/kubernetes

Envoy

https://www.envoyproxy.io/



https://github.com/envoyproxy/envoy

Envoy is an L7 proxy and communication bus designed for large modern service oriented architectures.

Originally developed at Lyft, Envoy is a self contained process that is designed to run alongside every application server. All of the Envoys form a transparent communication mesh in which each application sends and receives messages to and from localhost and is unaware of the network topology.

Serverless Framework

https://www.serverless.com/framework/docs



https://github.com/serverless/serverless

The Serverless Framework – Build applications on AWS Lambda and other next-gen cloud services, that auto-scale and only charge you when they run. This lowers the total cost of running and operating your apps, enabling you to build more and manage less.

The Serverless Framework is a command-line tool with an easy and approachable YAML syntax to deploy both your code and cloud infrastructure needed to make tons of serverless application use-cases. It's a multi-language framework that supports Node.js, Typescript, Python, Go, Java, and more. It's also completely extensible via over 1,000 plugins which add more serverless use-cases and workflows to the Framework.

Ansible

https://docs.ansible.com/



https://github.com/ansible/ansible

Ansible is a radically simple IT automation system.

It handles configuration management, application deployment, cloud provisioning, ad-hoc task execution, network automation, and multi-node orchestration. Ansible makes complex changes like zero-downtime rolling updates with load balancers easy.